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Original article

Multi-level Hierarchical Structure Analysis of Influencing Factors of Coal Mine Managers' Blame Avoidance Behavior

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ABSTRACT

Background: In the Chinese coal industry, widespread blame avoidance behavior (BAB) greatly impacts coal mine accidents. Therefore, it is necessary to stop the BAB of coal mine managers and raise the management level of coal mine enterprises for the safe development of Chinese coal industry.
Methods: Based on the semi-structured interviews and questionnaire surveys (20 middle-level managers in coal mines), this paper used the Grounded Theory and Nvivo Software qualitative research methodology to open, spindle, and selectively encode the interview data. Then, an index system of factors influencing BAB of coal mine managers was constructed. The influence degree, affected degree, centrality degree, cause degree of each influencing factor were calculated and the hierarchical model of influencing factors of BAB of coal mine managers was established by the DEMATEL-ISM method which is a decision support tool used to evaluate and analyze the interdependencies between influencing factors.
Results: Index system of factors influencing BAB of coal mine managers included four levels and 12 influencing factors: individual, organizational, institutional environmental, and situational factors. The hierarchical model identified eight causal factors and four consequential factors, of which safety management (15.355), work attitude (14.380), and work group performance (14.281) in the top three of the centrality rankings are the key factors affecting the avoidance behavior of coal mine managers; A 3-level multilevel structure was constructed to reflect the interactions among the factors influencing the BAB of coal mine managers in terms of direct, indirect, and root causes, and corresponding improvement measures were proposed.
Conclusion: This study offers a theoretical complement and practical guidance for stopping BAB of coal mine managers.

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1. Introduction

The overall number of coal mine fatalities in China has been declining in recent years, and the number of fatalities in 2020 has reduced by 77.78% compared to 2008 [1]. However, Chinese coal mine accident fatality rate is still quite high when compared to developed countries, accounting for more than 70% of all accidents worldwide [2]. The situation for coal mine safety remains dire. According to accident data, human unsafe behavior is to blame for more than 90% of coal mine mishaps [3]. There are various causes for unsafe behavior [4]; it is attributed to the tendency of individuals' behavioral decisions to generate deviations [5,6]. Furthermore, due to the advancements in new technologies, there is an emergence of associated risks [7]. Therefore, analyzing accidents should not be limited to individual perspectives but should encompass a comprehensive consideration of the system and

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complexity within the operational environment [8,9]. It is imperative not only to ensure effective personnel behavior control but also to implement efficient organizational management to mitigate risks [10]. Coal mine managers as a part of management, their blame avoidance behavior (BAB) has a significant effect on coal mine safety production.

Western academics have studied BAB in two stages: initial and developing stages. R. Kent Weaver was the research representative in the preliminary stage. He identified eight different strategy types of BAB, from the controversy strategy such as "redefining issues" to shifting blame and accusations onto others through "looking for scapegoats" [11]. Markus Hinterleitner found that the best avoidance strategy should start with the accountability mechanism, policy, experience strategy, and context factors [12]. In order to lessen the loss of credibility and financial resources when the government is accused by the media and the public, Sten Hanson studied how the government employs discourse methods to rationally avoid blame [13]. Hood advocated classifying strategies of BAB into institutional, surface, and policy categories [14]. Harpal Hungin revealed that the Central Bank of the United Kingdom after the financial crisis avoided responsibility by using the authorization strategy during the reform [15]. At this stage, research has been primarily concerned with describing the occurrence of BAB, such as how the decision-making bodies avoid taking responsibility, but has not gone into great detail to examine its underlying behavioral logic.

In terms of research fields, research paradigms, and explanatory frameworks, the developmental phase that Christopher Hood represented was richer than the first phase [16,17]. The study extended from the political sphere to public policy and administration, involving partisan BAB in higher education earmarks [18], the collision between the idea of openness and the logic of BAB [17], BAB arising from policy formulation and policy innovation [19], political crises and natural disasters, etc. [20]. From descriptive research to interpretative research, the research paradigm has changed. The process and effects of BAB have been thoroughly examined in these studies in terms of explanatory frameworks, and a theoretical discussion has progressively begun.

The domestic research of BAB is mostly on management, sociology, and political science. The definition, description, internal reasoning, and major tactics of BAB are covered. For instance, in accordance with the different definitions of "responsibility", Ni Xing et al. [21] defined the concept of "BAB" as actions taken by managers with a view to evading accountability. The BAB was explained by Wang Rui et al. [22] from the standpoint of power and responsibility, and they thought that the risk served as a link between the structure of power and responsibility and the act of avoidance. The blame avoidance strategy of leadership team was explained by Zhang Minmin and Li Li [23], who suggested that the formation of BAB is strongly influenced not only by the subjective environment but also by the objective environment. The formation of avoidance behaviors is correlated with the manner and capacity of risk perception, according to the argument of Guo Weiqing et al. [24] that decision-makers will initially decide based on risk perception when producing avoidance behaviors. Peng Zongchao [25] classified blame avoidance strategies into four types based on two dimensions: the time when the avoidance behavior occurs and the attitude of the decision-maker.

To sum up, domestic and foreign research on BAB focused on the field of politics and grass-roots government, while the BAB of managers in Chinese coal mine enterprises is extremely common, which will inevitably lead to accidents if left unchecked. Therefore, this paper applies the BAB to coal mine management, analyzes the factors influencing the BAB by the grounded theory (GT) and the DEMATEL-ISM method, and puts forward the improvement measures in a targeted way, with a view to improving the safeguard role of the coal mine managers in the safety management.

2. Materials and methods

2.1. Research methodology and data collection

Given the diversity of elements that affect coal mine managers' BAB, there are disparities in personal knowledge as a result of the mixture of subjective and objective factors, which need to be compared and summarized in a step-by-step manner. As a result, GT, a qualitative research methodology that builds a socially tested theoretical system and conceptual framework on the basis of methodically arranging and analyzing raw data information, is the method of study for this paper. Although there is no single definition for the many variants of GT, the commonalities of most approaches are systematic, inductive, and comparative, and they aim to establish a theoretical framework that explains why and how an individual, organization, or community responds to an event, challenge, or problem [26]. The methodology used in this paper is a variation of Corbin's and Strauss's [27,28], which includes open coding, spindle coding, and selective coding.

DEMATEL (Decision Making Trial and Evaluation Laboratory) is a method to analyze system elements by using graph theory and matrix tools, and to analyze the logical relationship and direct influence relationship between various elements in the system. It can judge whether there is a relationship between the elements and evaluate its strength and weakness.ISM (Interpretative Structural Modelling) is characterized by decomposing a complex system into several subsystems (elements) and constructing the system into a multi-level hierarchical structural model through algebraic operation. Dematel-ISM model synthesizes the advantages of DEMATEL method and ISM method, can identify the key elements in the system and their influence degree, and construct the hierarchical structure of the elements. It is often used as a decision support tool to evaluate and analyze the interdependence between influencing factors.

The interview outline was developed based on the pertinent research findings from domestic and international sources. To learn how they saw BAB in their day-to-day work as well as their impressions and opinions for the pre-interview outline, five coal mining managers were chosen for preinterviews. Finally, after combining with the responses and ideas from the participants who took part in the preinterviews, mature interview questions were created, and 20 middle-level coal mine managers were selected to participate in the official interviews.

Importing text data into the NVivo 12.0 software (QSR International Pty Ltd, Version 12, 2018) for processing analysis will help create three-level coding for dimensional creation and model identification of the BAB influencing factors for coal mine managers.

2.2. Data analysis

2.2.1. Open coding analysis

Line-by-line coding is a component of open coding that identifies the notion, its properties, and its dimensions. To begin with, the original data are decomposed and assigned a descriptive label, also known as a code. These codes are then extracted and compared with other codes to determine any similarities or differences. Codes with similar meanings are linked together, and they are organized into a concept that has related features if they have common features. The theoretical basis is made up of these ideas. The original text data are encoded and interpreted in accordance with the research topic, and a total of 255 effective initial concepts are

obtained. A total of 118 subcategories are created after the initial concepts are associated and clustered.

2.2.2. Spindle coding analysis

Data analysis has changed to concentrate on spindle coding, which involves reassembling data and figuring out connections between open codes, as a result of the identification of essential concepts. Spindle coding uses the concept established during the open coding period and categorizes it to give a more thorough explanation of the variables that could have an impact on coal mine managers' BAB. The data are continuously compared with the preceding data set during the spindle coding process. As a consequence of ongoing investigation and correlation of independent categories, there are 27 primary categories, including physical characteristics, psychological characteristics, moral emotions, and work attitudes, as shown in Table 1.

2.2.3. Selection coding analysis

Selective coding is a deepening of open coding and spindle coding, which aims to clarify and integrate existing categories and build their internal connections and logical structures. In this stage, the key theme is established as the core concept, and the low-level categories are integrated and arranged to explain the relationship between different categories of variables. Through the comparison and verification of the original sentences, the four core categories of individual factors, organizational factors, institutional environmental factors, and situational factors are finally determined. The results of selective coding are shown in Table 2.

2.2.4. Theoretical saturation test

Through the theoretical saturation test, the GT typically confirms the dependability and consistency of the three-level coding analysis results. A crucial stage of the GT analysis is the theoretical saturation test. The remaining five original text data are coded and examined in this work. It is demonstrated that this study's theoretical analysis has reached saturation when the findings of the coding analysis show no new concepts or categories.

2.3. Extraction of influencing factors library

After GT coding the original text data and combining it with the study's focus on coal mine managers' BAB, it was discovered that the four dimensions of individual, organizational, institutional environmental, and situational factors will all have an impact on coal mine managers' BAB. The individual factors include physiological characteristics, psychological characteristics,

Table 1
Spindle coding of coal mine managers' BAB

Serial number	Subcategory	Principal category
1	Genes, gender, age, size, appearance, and physical impairments	Physiological characteristics
2	Emotional state, volition, personality characteristics, disposition, character, and mode of thinking	Psychological characteristics
3	Moral sense, responsibility, and professional quality	Moral emotion
4	Social risk environment and enterprise internal environmental risk	Environmental risks
5	Serious, dedicated, responsible, energy conservation, risk assumption, mental state, psychological instinct, fortune, and nature	Work attitude
6	Memory, action, attention, reaction, and observation	Cognitive competence
7	Affective control	Affective management ability
8	Pressure catharsis ability, pressure control ability, and pressure resistance ability	Stress coping ability
9	Safety cognition, professional level, professional recognition, and professional mastery	Professional competence
10	Risk estimation, risk judgment, and risk experience	Risk perception ability
11	Environmental adaptability	Environmental adaptability
12	Personality insult	Traumatic experience
13	Self-confidence, behavioral cognition, self-happiness perception, self-authenticity manifestation, social adversity perception, and intimacy perception	Self-cognitive ability
14	Safety attitude, rule recognition, work values, work enthusiasm, work satisfaction, organizational belonging, work familiarity, cognitive bias, and cognitive inhibition	Occupational cognitive ability
15	Social barriers, social response, gaze direction, tone, and facial expressions	Social skills
16	Position, job position, job type, work station, working years, and working hours	Operational environment
17	Urban-rural differences, stability of educational environment, stability of educational environment, parenting style, partner situation, family composition, family economic status, personal income level, family pressure, work pressure, role pressure, and social pressure	Living environment
18	Family conflicts and domestic calamity	Family performance
19	Colleague behavior, colleague attitude, leadership behavior, leadership attitude, leadership character, leadership style, and leadership ability	Work organization performance
20	Family support, friend support, work organization support, work environment support, promotion mechanism support, social recognition, social class, state support, and media coverage	Organizational support
21	Safety rules and regulations, safety input, safety atmosphere, corporate culture, and protective measures	Safety management
22	Enterprise management style, enterprise attributes, work system, organizational justice, supervision and assessment mechanism, and responsibility division mechanism	Daily management
23	Lack of incentives, reward, and punishment mechanism	Incentive mechanism
24	Corporate profitability, corporate reputation, and advanced level of equipment	Production operation management
25	Accountability overload	Accountability mechanism
26	Fault tolerance and error correction management	Fault tolerance and error correction mechanism
27	The matching degree of power and responsibility	Power and responsibility distribution system

Table 2
Selective coding of coal mine managers' BAB

Serial number	Principal category	Core category
1	Physiological characteristics	Individual factors
2	Psychological characteristics	
3	Moral emotion	
4	Work attitude	
5	Cognitive competence	
6	Affective management ability	
7	Stress coping ability	
8	Professional competence	
9	Risk perception ability	
10	Environmental adaptability	
11	Traumatic experience	
12	Self-cognitive ability	
13	Occupational cognitive ability	
14	Social skills	
15	Operational environment	Organizational factors
16	Work organization performance	
17	Organizational support	
18	Safety management	Institutional environmental factors
19	Daily management	
20	Incentive mechanism	
21	Production operation management	
22	Accountability mechanism	
23	Fault tolerance and error correction mechanism	
24	Power and responsibility distribution system	
25	Environmental risks	Situational factors
26	Living environment	
27	Family performance	

moral emotion, work attitude, cognitive competence, affective management ability, stress coping ability, professional competence, risk perception ability, environmental adaptability, traumatic experience, self-cognitive ability, occupational cognitive ability, and social skills. There are three organizational factors: operational environment, work organization performance, and organizational support. Seven aspects make up the institutional structure factors: safety management, daily management, incentive mechanism, production and operation management, accountability system, fault tolerance and error correction mechanism, and power and responsibility distribution system. Situational factors include environmental risks, living environment, and family performance. The influencing factor model is drawn as shown in Fig. 1.

2.4. Construction of influencing factor index system

Using the GT approach, we identified 27 influencing factors on coal mine managers' BAB. It is vital to optimize the indicators for preventing overlap between them. In order to filter out more accurate judgments on the influencing variables, the questionnaire survey is used to gather and integrate expert judgment. The importance of the indicators was utilized to rank the various influencing elements.

The selection of experts should be based on their expertise in coal mine management due to the particularity of the frequent occurrence of emergencies in the coal mining industry and the complexity of middle managers' duties. In this study, a total of 15 experts were chosen (5 in the first round and 10 in the second), including technical, ventilation, electromechanical, safety, and production management personnel from coal mines, as well as external experts. These experts had an average age of 33.8 ± 4 years and an average working life of 8.13 ± 6 years. Following the completion of the survey, the mean, standard deviation, and coefficient of variation were computed. Fig. 2 displays the results of the final computation.

The screening results for the influencing factor indicators should be chosen based on the significance of the indicator and the greater level of expert coordination, and the indexes with a mean value above 2.5 and a coefficient of variation below 0.25 should be chosen. From the three domains of subjective factors, environmental factors, and institutional structure factors, twelve influencing elements were ultimately chosen. In conclusion, Table 3 depicts the index system of influencing factors for coal mine managers' BAB.

2.5. Establishment of hierarchical model of influencing factors

2.5.1. Method and model

There are certain similarities between DEMATEL and ISM that can be utilized to analyze and make decisions about complex systems [29]. ISM and DEMATEL were chosen as the dominant and auxiliary models in this investigation, respectively.

2.5.2. DEMATEL-ISM method calculation steps

The following steps of the DEMATEL-ISM approach are used to investigate the variables affecting the BAB of coal mine managers:

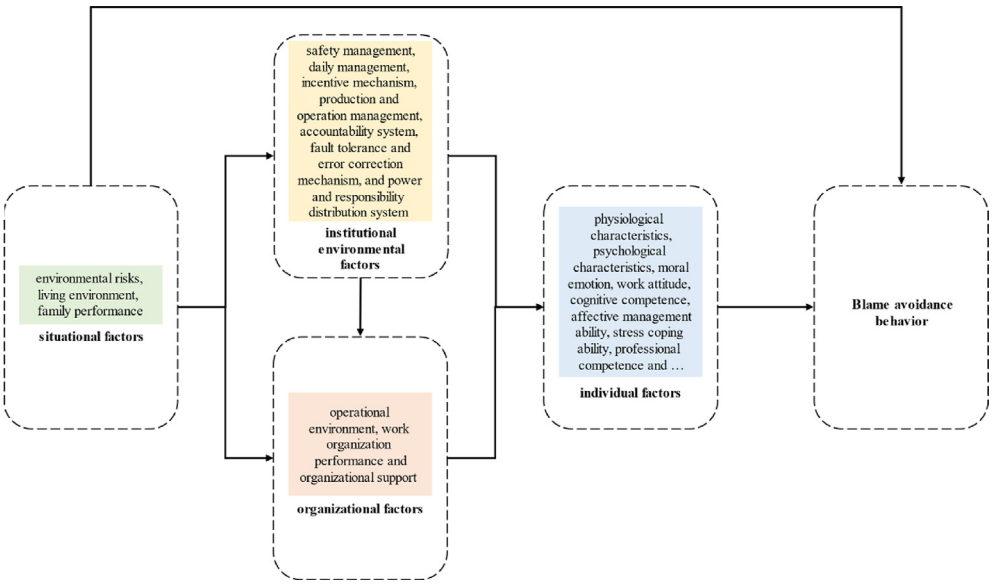


Fig. 1. GT of influencing factors of coal mine managers' BAB. BAB, blame avoidance behavior; GT, grounded theory.

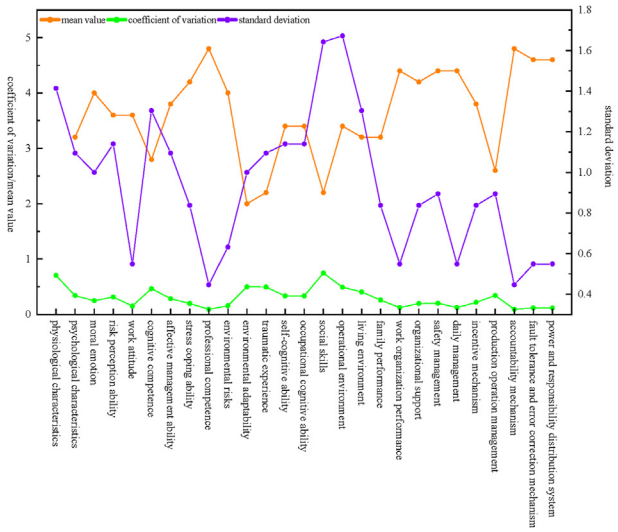


Fig. 2. Original influencing factors of coal mine managers' BAB and statistical analysis results of expert scoring. BAB, blame avoidance behavior.

Table 3
Index system of influencing factors of coal mine managers' BAB

First-level indicators	Second-level indicators	Third-level indicators
Index system of influencing factors of BAB	Individual factors	Working attitude (X1)
		Stress coping ability (X2)
		Professional ability (X3)
	Situational factors	Environmental risks (X4)
	Organizational factors	Work group performance (X5)
		Organizational support (X6)
	Institutional environmental factors	Safety management (X7)
		Daily management (X8)
		Incentive strategy (X9)
		Accountability system (X10)
		Fault tolerance and error correction mechanism(X11)
		Power and responsibility distribution system (X12)

BAB, blame avoidance behavior.

(1) Determining factors influencing the BAB of coal mine managers X1, X2 … Xn.
(2) Building the direct influence matrix. An n × n matrix is constructed from the average of the experts' scores, i.e., the direct influence matrix F.
(3) Building the comprehensive influence matrix. The comprehensive relationship between multiple elements, including direct and indirect relationships, can be reflected in the comprehensive influence matrix. Using formula (1), the comprehensive influence matrix T [30] is produced.

$$T = Y(I - Y)^{-1} \tag{1}$$

where $(I–Y)^{-1}$ is the inverse matrix of (I–Y) and I is the identity matrix.

(4) Calculation and analysis of the influence factor index. The influence factor index is determined using the data in the comprehensive influence matrix T and the formulas (2) – (5).
① Influence degree: The sum of the factors in each row of matrix T represents the comprehensive influence value of the corresponding factors on other factors in the row. It is expressed by D, and the calculation formula is

$$D_i = \sum_{j=1}^{n} t_{ij} \tag{2}$$

② Affected degree: The sum of factors in each column of matrix T represents the comprehensive influence value of all factors in the column by other factors. It is expressed by R, and the calculation formula is

$$R_i = \sum_{i=1}^{n} t_{ij} \tag{3}$$

③ Cause degree: Referring to the influence degree minus the affected degree, that is, D_i - R_i.

$$E_i = D_i - R_i \tag{4}$$

④ Centrality: Referring to the sum of the influence degree and the affected degree, namely $D_i + R_i$.

$$M_i = D_i + R_i \tag{5}$$

(5) Drawing a causality diagram where the ordinate and abscissa indicate the cause degree E_i and the center degree M_i, respectively. Determining the causal attributes of the factors and identifying the key factors that influence BAB.
(6) Building the overall influence matrix Z. the formula is

$$Z = T + I \tag{6}$$

(7) Determining the threshold λ and the reachability matrix. On the basis of the overall influence matrix Z, the threshold is determined by the formula $\lambda = \alpha + \beta$ (where α and β are the average and standard deviation of the elements in the comprehensive influence matrix, respectively), and then the reachability matrix K is obtained according to Formula (7).

$$\begin{cases} K_{ij} = 1 \; Z_{ij} \geq \lambda \\ K_{ij} = 0 \; Z_{ij} < \lambda \end{cases} \tag{7}$$

(8) Determining the reachable set R_i, the antecedent set A_i, and the common set O_i of factor X_i. Through the reachable matrix K, the reachable set R_i and the antecedent set A_i are obtained, where R_i is the set of i row factors containing element 1 in the reachable matrix K, A_i is the set of j column factors containing element 1 in the reachable matrix K, and the expression is

$$R_i = \{x_i | x_j \in X, K_{ij} \neq 0\} \tag{8}$$

$$A_i = \{x_i | x_j \in X, K_{ij} \neq 0\} \tag{9}$$

$$Q_i = R_i \cap A_i \tag{10}$$

(9) Dividing the rating of influencing factors extracts the same influencing factors contained in the reachable set R_i and the common set O_i; that is, X_i is at the highest level in the system. After extracting X_i, the i-th row and i-th column are deleted in the matrix K, and the method is repeated until all factor ratings are divided.

3. Results

The remaining 10 experts were asked to rate the 12 influencing factors according to the 0–4 scoring method, wherein 4, 3, 2, 1, and 0 are high impact, higher impact, medium impact, low impact, and no impact, respectively, based on the index system of the influencing factors of the BAB of coal mine managers in Table 1. The average value of 10 score tables is calculated to get the direct influence matrix F. Using formula (1), the comprehensive influence matrix T is produced, as illustrated in Table 4.

Based on the data in the comprehensive influence matrix T, the indicators of factors influencing the BAB are calculated according to Eqs.(2)–(5), and the related DEMATEL analysis results are shown in Table 5, and then the horizontal coordinate is used to indicate the centrality M_i and the vertical coordinate to indicate the cause degree E_i, and then the causality diagram is drawn, as shown in Fig. 3.

3.1. Cause degree analysis

Important indicators of factor attributes are classified by cause degree. When the degree of cause is greater than 0 (i.e., above the X axis), that is, the factor of cause, it means that it has a greater impact on other factors. The factor is more affected by other factors when the cause degree is less than 0 (i.e., below the X axis), which is the result factor, and the cause factor will influence the result factor (via Fig. 3). X4, X6, X7, X8, X9, X10, X11, and X12 are the cause factors. It indicates that these eight factors are susceptible to interference with other factors and that in controlling the influence factors, attention is needed and the transfer process between factors is blocked. The four influencing elements in the diagram are X1, X2, X3, and X5. These four factors are the result factors and are positioned below the X-axis, suggesting that they are susceptible to influence by other factors. It should be emphasized that when controlling the affecting factors, these factors are not interfered with.

3.2. Centrality analysis

The centrality indicates the role of an element in the system. The graph's distribution is more right-sided and has a higher value when an element is more significant. The top three influencing factors of the centrality ranking in Fig. 3 are X7, X1, and X5, indicating that these three variables are most likely to have a direct impact on the occurrence of BAB in coal mine managers. Consequently, X7, X1, and X5 are chosen as the key influencing elements.

Table 5
Indicators of influencing factors of coal mine managers ' BAB

Factor	Influence degree (D)	Affected degree (R)	Centrality (M)	Cause degree (E)	Centrality ranking	Factor attribute
X1	6.281	8.099	14.380	-1.819	2	Result
X2	5.599	6.349	11.948	-0.751	12	Result
X3	6.333	6.337	12.671	-0.004	10	Result
X4	6.753	6.400	13.153	0.353	9	Cause
X5	6.813	7.468	14.281	-0.656	3	Result
X6	6.114	6.107	12.221	0.007	11	Cause
X7	7.726	7.628	15.355	0.098	1	Cause
X8	6.829	6.794	13.623	0.035	5	Cause
X9	6.963	6.571	13.534	0.392	7	Cause
X10	7.276	6.569	13.845	0.706	4	Cause
X11	7.026	6.578	13.604	0.449	6	Cause
X12	7.315	6.126	13.440	1.189	8	Cause

BAB, blame avoidance behavior.

3.3. The establishment of hierarchical model

Determining the threshold λ is essential to creating the reachability matrix. If one factor has an impact on another that is greater than λ, the stimulus will be responded to, meaning that the factor can directly affect the other factor; if one factor has a lesser influence over another that is less than λ, the stimulus will not be responded to, meaning that the factor has no influence over the other factor. The goal of thresholding, which is also the core of Leveson's causal composite model [31], is to emphasize the primary influencing aspects of complex systems without oversimplifying them. Using the formula (7) to determine the reachability matrix K and calculate the threshold $\lambda = 0.636$, then combining with formula (8)–(10) to divided into different levels of each influencing factor and draw a hierarchical structure diagram accordingly. In this paper, the hierarchical structure of influencing factors of BAB is subdivided into direct, indirect, and fundamental factors, as shown in Fig. 4.

The fundamental rating L1 is the direct factor, which includes X1, X2, and X6. The occurrence of coal mine managers' BAB is most directly influenced by these three variables. X3, X5, X7, X8, X9, X10,

Table 4
The comprehensive influence matrix describes the comprehensive relationship between the factors affecting the BAB of coal mine managers

Factor	X1	X2	X3	X4	X5	X6	X7	X8	X9	X10	X11	X12
X1	0.553	0.519	0.520	0.524	0.594	0.476	0.578	0.521	0.507	0.506	0.507	0.475
X2	0.578	0.387	0.473	0.449	0.512	0.430	0.521	0.471	0.458	0.458	0.458	0.403
X3	0.644	0.497	0.437	0.527	0.573	0.480	0.584	0.526	0.537	0.537	0.511	0.480
X4	0.707	0.525	0.551	0.472	0.634	0.534	0.644	0.556	0.541	0.541	0.541	0.508
X5	0.686	0.529	0.529	0.560	0.554	0.512	0.650	0.589	0.572	0.545	0.573	0.513
X6	0.624	0.481	0.480	0.485	0.582	0.406	0.592	0.510	0.496	0.496	0.496	0.466
X7	0.766	0.593	0.590	0.623	0.714	0.599	0.643	0.657	0.637	0.664	0.639	0.602
X8	0.686	0.531	0.529	0.534	0.641	0.512	0.652	0.506	0.574	0.547	0.575	0.541
X9	0.698	0.566	0.566	0.544	0.650	0.548	0.661	0.599	0.498	0.556	0.556	0.522
X10	0.725	0.588	0.559	0.566	0.677	0.542	0.716	0.624	0.579	0.521	0.608	0.572
X11	0.704	0.571	0.542	0.548	0.656	0.524	0.669	0.605	0.561	0.588	0.504	0.554
X12	0.729	0.564	0.562	0.568	0.681	0.545	0.720	0.628	0.609	0.610	0.610	0.489

BAB, blame avoidance behavior.

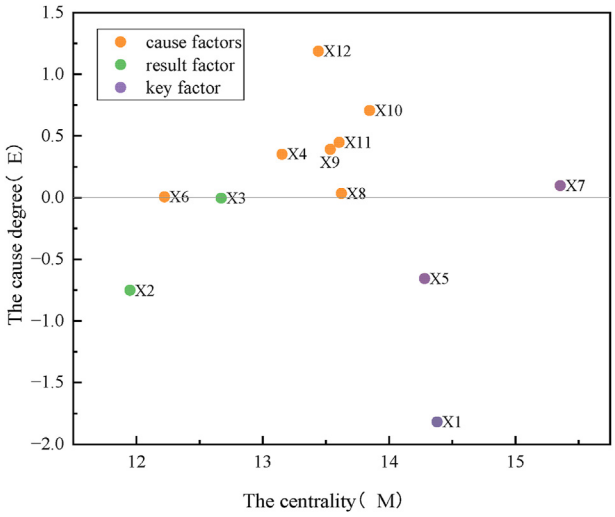


Fig. 3. Causal relationship diagram of influencing factors of coal mine managers ' BAB. BAB, blame avoidance behavior.

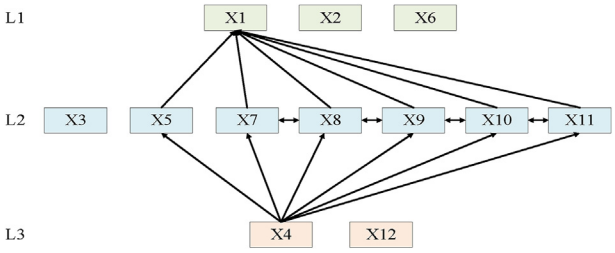


Fig. 4. Hierarchical model of influencing factors of coal mine managers ' BAB. BAB, blame avoidance behavior.

X11, and the basic rating L2 are indirect factors that connect these factors. The fundamental factor is the basic rating L3, which also includes X4 and X12.

4. Discussion

BAB is a common occurrence in coal mine accident cases, and it is impossible to overlook how this risky behavior affects the safety management of coal mine firms. In order to investigate the influence mechanism of coal mine managers' BAB, this article combines GT and DEMATEL-ISM methodologies.

First of all, the individual factors, situational factors, organizational factors, and institutional environmental factors are the four factors that affect the BAB of coal mine managers. Individual factors like their own behaviors. People tend to make judgments during the process of behavioral decisions based on their own cognition and motivation [32]. Work attitude, stress coping ability, and professional ability are the subject factors. Among the situational factors, the complexity and system of environmental risks should be considered in specific situations [33]. Work organization performance and organizational support are examples of organizational influence; when employees feel supported by the company, they are more likely to work more effectively and have fewer BAB incidents in addition to feeling the company's love, care, and appreciation [34]. The institutional structural factors include safety management, daily management, incentive strategy, accountability system, fault tolerance and error correction mechanisms, and power and responsibility distribution systems.

Secondly, safety management, work attitude, and work group performance are the three most important centrality variables, indicating that these three elements can easily cause the occurrence of coal mine managers' BAB, which is the key to stopping BAB. Safety management's importance comes in front. It is simple for managers to evade obligations in the form of loopholes if the organizational safety management work is not put in place [35]. The second is the attitude at work. Managers develop a specific work attitude based on their prior experiences and cognitive processes [36]. When faced with hazards, they will determine whether they are capable of carrying out their responsibilities as "economic men." The third factor is how the work organization is performing. People are subjected to the influences like human interactions and relationships [37]. Managers will unwittingly shirk their duties due to their coworkers' BAB and a poor work environment.

Finally, there are three levels into which the influencing variables of BAB can be subdivided: direct factors, indirect factors, and fundamental factors. The power and responsibility distribution system as well as professional cognitive ability are the root variables. As a result, strengthening the professional cognition of each individual employee as well as the system of power and responsibility allocation inside the company is the primary remedy to BAB.

The results of this study show that avoidance behavior is a complex social phenomenon, which needs to be studied and intervened from multiple perspectives. In order to reduce the occurrence of avoidance behavior, it is necessary for managers and employees at all levels to work together to strengthen safety education, improve system construction, strengthen supervision, and build a good corporate culture. At the same time, it is also necessary for guide managers to correctly understand and evaluate the avoidance behavior. Once a significant violation is identified, an exhaustive investigation into the causes is requisite, the loopholes in organizational management are found out, and then efficacious countermeasures are formulated and implemented. Furthermore, education, training, and assessment should be carried out for the party and other managers who may encounter similar problems so as to create a good atmosphere for the participation of the whole enterprise.

CRediT authorship contribution statement

Li Wang: Funding acquisition, Project administration, Supervision, Conceptualization. **Wenyi Zeng:** Software, Writing – original draft, Writing – review and editing, Formal analysis, Investigation, Methodology, Visualization. **Ziyan Jing:** Data curation, Investigation, Methodology, Resources, Software, Writing – original draft. **Yonghui Xu:** Software, Validation, Visualization. **Jiang Li:** Conceptualization, Validation, Visualization.

Conflicts of interest

Author states that there is no conflict of interest.

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